

Lab: Thought Experiment — Learning, Transformation, and Niche Construction

Objective

Investigate the philosophical implications of learning as self-transformation and niche construction through structured thought experiments.

Part 1: The Apprentice's Transformation

Goal: Analyze learning as self-transformation, not mere information acquisition.

Consider a medical student who, after years of training, can look at a patient and immediately recognize the signs of a disease invisible to the untrained eye.

- Has the student acquired new *information* or become a new *kind of perceiver*? What is the philosophical difference?
- How does Active Inference formalize this transformation (hint: generative model restructuring, precision changes)?
- Does the student who graduated remember what it was like *not* to see the disease? What does this suggest about the nature of learning?

Write your response here

Part 2: Niche Construction — The Built Environment

Goal: Explore how agents transform their environments to reduce surprise.

Humans build houses, write books, create institutions, and establish laws. Each of these is a form of niche construction — restructuring the environment to make it more predictable.

- Choose one human niche construction activity (e.g., writing a calendar, building a road, creating a law). Analyze it as an Active Inference strategy: how does it reduce expected surprise?
- Is niche construction "cheating" — avoiding the need to learn by making the environment simpler? Or is it the most sophisticated form of learning?
- How does this relate to Clark's concept of "cognitive scaffolding"?

Write your response here

Part 3: The Paradigm Shift

Goal: Apply structure learning to revolutionary changes in worldview.

Kuhn argued that scientific revolutions involve not merely new data but a change in the entire framework for interpreting data — a "paradigm shift." Before and after the shift, scientists literally "see" different things.

- How does Bayesian Model Reduction (BMR) — comparing nested models and selecting the one with best evidence — relate to Kuhn's account?
- Is a paradigm shift more like parameter learning (refining existing predictions) or structure learning (changing the model itself)?
- Can BMR explain why paradigm shifts are often resisted? What does "resistance to learning" look like in Active Inference terms?

Write your response here

Part 4: Personal Identity Under Transformation

Goal: Engage with the philosophical problem of identity through change.

If I am my generative model, and my generative model changes through learning, am I still the same person? Consider the case of a profound life experience (e.g., becoming a parent, surviving a trauma) that fundamentally restructures one's priorities and perceptions.

- Is the post-learning agent the "same" agent as the pre-learning one?
- How does this relate to Parfit's concept of personal identity as a matter of psychological continuity rather than substance identity?
- Write a 100-word reflection on a learning experience that changed who you are, not just what you know.

Write your response here

Part 5: Synthesis and Position Defense

Goal: Articulate your philosophical position on learning.

Write a 200-word statement on one of:

- "Learning is the most philosophically radical concept in Active Inference because it implies the agent's identity is always in flux."
- "Niche construction is just as important as perception and action — and philosophy has neglected it."
- "Bayesian Model Reduction captures the logic of paradigm shifts better than Kuhn's own account."

Share with a peer and defend against their strongest objection.

Write your response here

Lab Summary

Part	Skill Developed	Key Philosophical Move
1	Transformation analysis	Distinguishing information acquisition from perceptual transformation
2	Environmental analysis	Analyzing niche construction as a cognitive strategy
3	History of science reasoning	Connecting structure learning to paradigm shifts
4	Identity reasoning	Examining personal identity under transformative learning
5	Position defense	Constructing and defending a philosophical position on learning